MEDICARE · DRUG PRICING · INNOVATION POLICY

The Innovation Clock

When do America's most expensive Medicare drugs earn their clinical value — and how does that timing collide with the Inflation Reduction Act's price-negotiation deadline?

20 of 30

negotiated small-molecule drugs gained a brand-new FDA-approved indication during their year 7–9 window — the very years the IRA's negotiation clock cuts short for pills. Separately, **25.4% of new indications** across the 40 negotiated drugs arrived *after* the drug's negotiation clock would already be running. Those drugs account for ~**$112.2B** in gross Medicare spending (Part D + Part B, 2023).

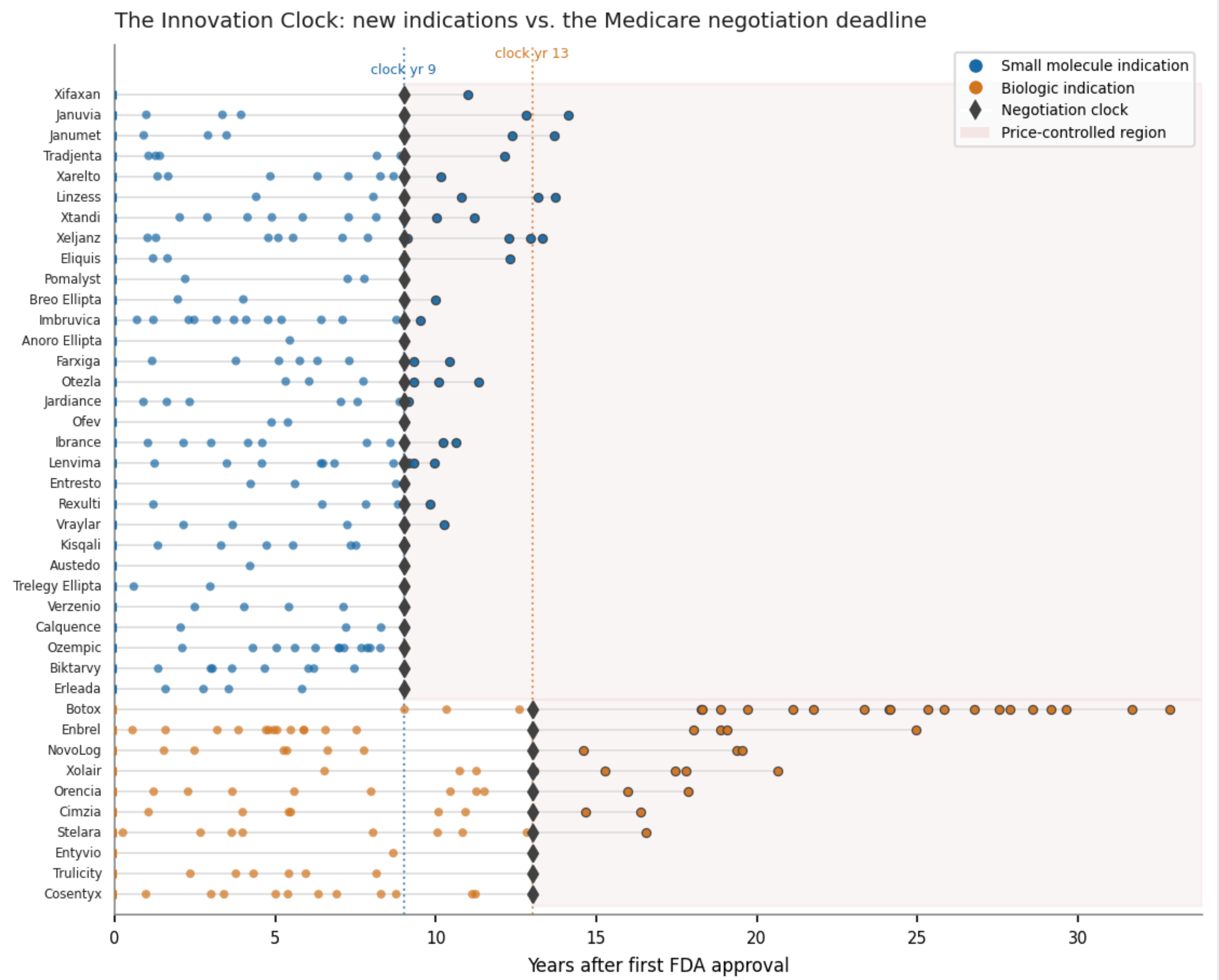

Figure 1. Each lane is an IRA-negotiated drug, from first FDA approval (year 0) to its most recent new indication. Dots are new indications (FDA efficacy supplements); the diamond is the negotiation clock — year 9 for small molecules (NDA), year 13 for biologics (BLA). The shaded band is the period when Medicare's negotiated price is in effect. Biologics keep earning indications years past the point where small molecules are already price-controlled.

The IRA lets Medicare negotiate prices for its highest-spend, single-source drugs. But the clock runs differently by **modality**: a small-molecule pill (FDA "NDA") can be selected at year 7 with a negotiated price at year 9, while a biologic ("BLA") is shielded until year 11/13. [1] That four-year gap is the contested "**pill penalty**." Why does timing matter for value? Because drugs rarely arrive fully formed — they accumulate new approved uses for years. In this cohort, small-molecule indications land a median of **5.84 years** after launch, and **44.3%** arrive in the back half of the pre-clock window. Roughly **19.2%** of small-molecule indications — and **35.6%** of biologic indications — are approved only *after* the negotiation clock has struck. The policy question is whether negotiating a pill's price at year 9 chills the late-stage research that produces those additional indications — or whether that fear is, as critics put it, overstated.

Spend, longevity, and the modality split

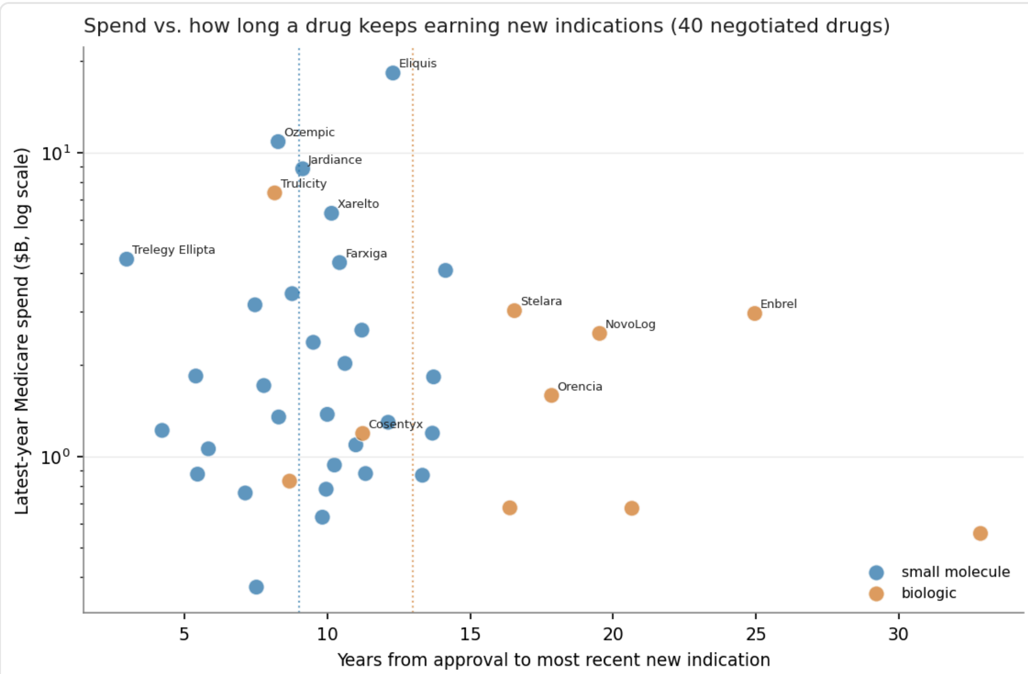

Figure 2. Latest-year Medicare spend vs. how long each of the **40 negotiated drugs** keeps earning new indications. Biologics (amber) cluster to the right — they keep expanding well past year 13 — while the highest-spend small molecules (e.g. Eliquis, Ozempic, Jardiance) reach their negotiation clock at year 9.

Table. New indications by modality across the 40 negotiated drugs (# = efficacy-supplement events; 30 small molecules + 10 biologics).

MODALITY	# NEW INDICATIONS	MEDIAN YRS TO INDICATION	% IN BACK HALF OF WINDOW	% AFTER THE CLOCK
Small molecule — clock yr 9	167	5.84	44.3%	19.2%
Biologic — clock yr 13	101	9.03	26.7%	35.6%

Rare-disease angle

Of the negotiated cohort, **19** drugs carry at least one FDA-approved orphan (rare-disease) indication, and **12** carry two or more. The **One Big Beautiful Bill Act** (signed July 4, 2025) now exempts "serial orphan" drugs — multiple rare-disease indications and *no* non-orphan approval — and resets their negotiation clock to first non-orphan approval. [3] Applying that stricter test (multiple orphan indications *and* no approved non-orphan indication), **4** negotiated drugs (Calquence, Imbruvica, Ofev, Pomalyst) plausibly qualify — Lenvima is excluded because it also treats renal cell and endometrial carcinoma (non-orphan). CBO estimated the expanded carve-out will cost Medicare roughly **$8.8B over 10 years (revised; ~$4.9B at enactment)**. [4]

What the data shows · what remains contested

What the data shows

- **20 of 30** negotiated small-molecule drugs gained a new indication in years 7–9 — pills *do* keep earning clinical value as the clock closes.
- Biologics keep approving new uses far longer: **35.6%** of their indications land after year 13, vs **19.2%** for pills after year 9.
- **Independent corroboration:** a peer-reviewed NPC study (Patterson et al., 2024) found **25% of oncology drugs** got their most recent subsequent indication only after they would be negotiation-eligible — closely matching this cohort's own **~25.4%**. [9]
- The 40 negotiated drugs account for **~$112.2B** in gross Medicare spending (Part D + Part B, 2023), *computed from CMS Spending-by-Drug data* — not CMS's official per-cycle totals, which use different 12-month windows.
- An EPIC-style 11-year rule moves *only* the small-molecule clock — biologics already sit at year 11. Under it, Medicare could **not** have negotiated **19 of the 30** negotiated small molecules — about **$67.6B** in spend. [2]

What remains contested

- **Industry (PhRMA/PHAR, 2023):** 61% of the 31 cancer drugs approved 2006–2012 gained at least one new indication after launch, and ~40% of those arrived 7+ years out — inside the window the year-9 clock truncates. [7]
- **Industry projection:** a University of Chicago analysis projects roughly **188 fewer small-molecule medicines** as a result of the 9-vs-13 gap. [8]
- **Critics (Public Citizen):** call the penalty a "myth," noting small-molecule venture investment has not fallen since the program passed. [5] An EPIC delay, they find, would have blocked 5 of 10 (>$37B); 8 of 15 (>$28B) drugs. [6]
- **Independent (KFF):** drugs that would be ineligible under a delayed (biologic-style) timeline accounted for about two-thirds of Part D spending on the first 25 selected drugs — **$61B of $91B**. [10]

Bottom line. The data cut both ways. Small molecules genuinely keep earning new indications inside the window the clock truncates — the kernel of truth in the innovation-incentive argument. Yet delaying selection to equalize with biologics would also pull tens of billions of dollars of high-spend drugs out of negotiation. Whether the EPIC Act is an innovation fix or a windfall depends on which of these effects you weight — and the same dataset supports both readings.

Sources & methodology

[1] **Congressional Research Service** — https://www.congress.gov/crs-product/R47872

[2] **U.S. Congress (Congress.gov)** — https://www.congress.gov/bill/119th-congress/house-bill/1492

[3] **Sidley Austin LLP** — https://www.sidley.com/en/insights/newsupdates/2025/07/key-inflation-reduction-act-amendment-broadens-us-protection-for-orphan-drugs

[4] **Congressional Budget Office** — https://www.cbo.gov/publication/61818

[5] **Public Citizen** — https://www.citizen.org/article/epic-act-would-be-a-multi-billion-dollar-pharma-windfall-at-the-expense-of-seniors/

[6] **Public Citizen** — https://www.citizen.org/article/epic-act-would-be-a-multi-billion-dollar-pharma-windfall-at-the-expense-of-seniors/

[7] **PhRMA (Partnership for Health Analytic Research)** — https://www.prnewswire.com/news-releases/new-analysis-inflation-reduction-act-undermines-cancer-medicine-development-301840713.html

[8] **University of Chicago analysis (cited via Epstein Becker Green)** — https://www.healthlawadvisor.com/medicare-drug-price-negotiation-program-the-inflation-reduction-act-pill-penalty-and-other-ira-reforms-on-the-horizon-for-2026

[9] **National Pharmaceutical Council (Patterson et al.)** — https://www.npcnow.org/resources/ira-may-have-unintended-consequences-subsequent-indications-oncology-drugs-new-research

[10] **KFF (Kaiser Family Foundation)** — https://www.kff.org/medicare/the-effect-of-delaying-the-selection-of-small-molecule-drugs-for-medicare-drug-price-negotiation/

[data] **CMS Drug Price Negotiation** — https://www.cms.gov/inflation-reduction-act-and-medicare/medicare-drug-price-negotiation

[data] **CMS Part D Spending by Drug** — https://data.cms.gov/summary-statistics-on-use-and-payments/medicare-medicaid-spending-by-drug/medicare-part-d-spending-by-drug

[data] **CMS Part B Spending by Drug** — https://data.cms.gov/summary-statistics-on-use-and-payments/medicare-medicaid-spending-by-drug/medicare-part-b-spending-by-drug

[data] **FDA Drugs@FDA / openFDA** — https://api.fda.gov/drug/drugsfda.json

[data] **FDA Orphan Drug Designations** — https://www.accessdata.fda.gov/scripts/opdlisting/oopd/

Cohort: all numbers and figures use the **40 IRA-negotiated drugs** (Cycles 1–3; 30 small molecule + 10 biologic). Modality = FDA application type of the earliest original approval (NDA = small molecule, BLA = biologic). "New indications" = FDA *efficacy supplements*, a documented proxy that can include some population/line-of-therapy expansions. **Spend** = gross Medicare spending (Part D + Part B, 2023) computed from CMS Spending-by-Drug data — not CMS's official per-cycle totals. **Orphan** status is from the real FDA Orphan Drug Designations database; the strict serial-orphan test is a documented heuristic. Cycle 3 Maximum Fair Prices are not yet published. Full pipeline + caveats: repository `docs/METHODOLOGY.md`.